

Source-Locked Occurrence–Phase Return Integration: Three Structural Roots, Metric All-Prefix Control, and the Named-Atom Firewall

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Abstract

TPC-163–170 open two more precise layers of the fixed-shift program. The frozen structural corpus contains 2988 production cut paths but no theorem-backed cut-to-actual-occurrence edge. Its 866 native triples identify neither rows nor occurrences: 2976 rows lie in collision classes, with excess 2122. The unique smallest archive key (ℓ, k, d, j_L, j_K) separates the frozen rows, but has addressing semantics only. A finite typed gluing theorem constructs a global formal row family from supplied compatible local families. It does not supply those families, actual active support, or a canonical minimal representation. The former monolithic crosswalk gap therefore splits into three independent not-testable roots.

The arithmetic branch proves Parseval, finite separated-registry, maximal all-prefix and Borel–Cantelli packet theorems on the actual determinant-two core. On an explicit packet schedule, every $\delta < 1/4$ is available for Lebesgue-almost every fixed phase. This is genuine L1 phase-metric progress, including every prefix in the prescribed terminal shells. It is not control at a named physical phase. An uncontrolled atomic registry may be concentrated on the schedule-dependent null set.

We integrate these results through a six-axis quantifier signature, separate complete architecture routes from arithmetic method subroutes, enforce an arithmetic-ancestor firewall on the physical registry, and introduce Endpoint Ledger V4. The selected map route has a seven-node minimal not-testable antichain. Its first reporting representative is the source-backed local occurrence-edge family. No positive program-L2, strict $1/400$, prime-pair lower bound, or twin-prime theorem is claimed.

1 Frozen structural state

Let $\mathcal{C}$ denote the frozen nonsoft cut archive. TPC-163 performs a source-locator census over the declared TPC-153, TPC-156, TPC-161 and TPC-162 corpus. It finds

$$|\mathcal{C}| = 2988, \quad \#\{\text{native triples}\} = 866, \quad (1)$$

and zero theorem-backed production edges in each required actual-occurrence class. This last statement is a census of one frozen corpus. It is not a theorem that actual occurrence edges do not exist.

The native triple is strongly noninjective:

$$2976 \text{ rows lie in collision classes,} \quad 2988 - 866 = 2122. \quad (2)$$

TPC-164 exhausts all nonempty subsets of the eight archived fields

$$\ell, k, d, j_L, j_K, D_0, \text{ reason, type.}$$

The unique smallest injective key is

$$K_{\text{arc}} = (\ell, k, d, j_L, j_K). \quad (3)$$

It gives a stable address for a row of this archive. It does not give an occurrence identifier, a physical parent, active support, or a canonical representation of an actual carrier.

Theorem 1.1 (Formal local-to-global gluing, imported from TPC-165). *Suppose a finite cover of $\mathcal{C}$ is supplied with nonempty finite typed local row families. Suppose further that the local families carry exact rational column sums equal to one and that their overlap bijections preserve all typed payloads and satisfy identity, inverse and triple-overlap composition. Then the quotient by the overlap relation is a global formal row family, unique up to the unique typed isomorphism commuting with the local maps. Exact column conservation descends to the quotient.*

The theorem is a finite descent statement. The production corpus does not currently supply a local family or overlap cocycle. Even when formal totality is supplied, two more logically independent predicates remain:

$$\begin{aligned} \mathbf{A} &= \text{every retained row has nonzero physical coefficient,} \\ \mathbf{M} &= \text{the actual carrier has a proved canonical or minimal representation.} \end{aligned} \tag{4}$$

Neither follows from [theorem 1.1](#), exact column conservation, or the archive address [\(3\)](#).

The source-backed local-edge target therefore carries the weaker support signature

SOURCE_BACKED_LOCAL_SUPPORT.

It is not assigned actual-active-support semantics. Only the later actual lift, which consumes the local-edge family together with both certificates in [\(4\)](#), has the support target

ACTUAL_ACTIVE_SUPPORT.

Proposition 1.2 (Refined crosswalk roots). *The selected occurrence-augmented map route has the independent minimal roots*

$$\begin{aligned} &\mathbf{H1.source_backed_local_occurrence_edge_family}, \\ &\mathbf{H1.actual_active_support_certificate}, \\ &\mathbf{H1.canonical_minimal_representation_certificate}. \end{aligned} \tag{5}$$

All three have status NOT-TESTABLE. The first is the selected reporting representative; this selection does not erase the other two.

Proof. The first predicate would instantiate the local families and their source locators in [theorem 1.1](#). The second concerns nonzero physical coefficients, and the third concerns representation minimality. The frozen source census, formal gluing theorem and archive address prove none of these implications. TPC-166 records them as parent-free production roots. Consequently none is an ancestor of another, proving minimality of the antichain. $\square$

2 The phase-metric arithmetic chain

For one determinant-two core cell write

$$c_z = \mu(d + sz)\mu(u + az), \quad su - ad = 2,$$

and on an interval I_N define

$$S_N(\alpha) = \frac{q}{N} \sum_{z \in I_N} c_z e(-\alpha z), \quad e(x) = e^{2\pi i x}. \tag{6}$$

TPC-167 observes the exact Parseval identity

$$\int_0^1 |S_N(\alpha)|^2 d\alpha = \frac{q^2}{N^2} \sum_{z \in I_N} |c_z|^2. \tag{7}$$

It applies at every scale and uses no Möbius cancellation. If $q \leq (\log X)^{\eta_0}$ and $N \geq \sqrt{X}$, its phase- L^2 envelope is $X^{-1/4}$ up to a polylogarithmic factor. The norm is analytic L^2 in phase, not program- L^2 .

TPC-168 proves the separated sampling inequality. For δ -separated $\alpha_1, \dots, \alpha_M$ and $P(\alpha) = \sum_{n < L} b_n e(n\alpha)$,

$$\sum_{j=1}^M |P(\alpha_j)|^2 \leq (\delta^{-1} + 4\pi(L-1)) \sum_{n < L} |b_n|^2. \quad (8)$$

This controls all but a sparse fraction of a sufficiently separated registry. It does not identify a distinguished element of that registry. In particular, density-one control cannot be evaluated at a named physical phase until the selector is supplied and justified.

TPC-169 replaces one terminal sum by every prefix in a fixed terminal shell. If

$$M_T(\alpha) = \max_{U \in [\theta T, T]} \left| \frac{q}{T} \sum_{z \leq U} c_z \rho(z) e(-\alpha z) \right|,$$

then dyadic orthogonality gives

$$\int_0^1 M_T(\alpha)^2 d\alpha \ll D_L^2 \|\rho\|_\infty^2 \left(\frac{q}{T} + \frac{q^2}{T^2} \right), \quad (9)$$

where $D_L \ll 1 + \log L$. Thus one phase-exceptional set controls every prefix in the declared shell. The normalization loss from $U \geq \theta T$ is explicit.

Finally, for a prescribed packet list $\mathcal{P}_n$, TPC-170 defines packet energies $V_{n,p}$ from the right side of (9). If

$$\sum_n \lambda_n^{-2} \sum_{p \in \mathcal{P}_n} V_{n,p} < \infty, \quad (10)$$

then Borel–Cantelli implies that for Lebesgue-almost every fixed α , all declared packets and all their prefixes are eventually at most λ_n . With $X_n = 2^n$, polylogarithmically many packets and moduli, $T_{n,p} \geq \sqrt{X_n}$, one may take

$$\lambda_n = X_n^{-\delta} \quad (\delta < 1/4). \quad (11)$$

Exact Abel summation also gives

$$\frac{q}{T} \left| \sum_j b_j e(-\alpha z_j) w_j \right| \leq M_T(\alpha) V(w). \quad (12)$$

Theorem 2.1 (Precise arithmetic advance). *Equations (7)–(12) prove an L1 actual-core phase-metric packet corridor, maximal over all prefixes in the prescribed shells. They do not prove a bound at a named fixed phase, at a scale-dependent phase selector, or on an unregistered physical packet.*

Proof. The positive conclusion is exactly (10). Its null set depends on the full prescribed packet schedule. An arbitrary atomic registry can concentrate all its mass on that null set. A constant-coefficient example at phase zero shows that no coefficient-blind metric theorem implies a named-atom theorem. This is a nonimplication for the uncontrolled atomic promotion method, not a lower bound for the literal Möbius correlation. $\square$

3 Six-axis quantifier contract

Every imported export and every DAG node carries

$$\mathbf{q} = (\mathbf{c}, \mathbf{p}, \mathbf{e}, \mathbf{s}, \mathbf{d}, \mathbf{a}), \quad (13)$$

where the coordinates record carrier, phase, endpoint, scale, decay and support quantifiers. The four arithmetic imports have signatures shown in table 1.

The promotion firewall forbids each of the following:

$$\begin{aligned} \text{Lebesgue-a.e.} &\implies \text{named fixed atom,} \\ \text{registry density} &\implies \text{distinguished registry atom,} \\ \text{formal support} &\implies \text{actual active support,} \\ \text{explicit schedule} &\implies \text{all deterministic scales.} \end{aligned} \quad (14)$$

These are logical quantifier changes, not losses that can be paid by shrinking an exponent.

Export	Carrier	Phase	Endpoint/scale	Decay/support
TPC-167	one actual core cell	Lebesgue L^2	terminal interval, every declared scale	phase-averaged X -power, actual core
TPC-168	separated finite registry	registry density	terminal interval, every declared scale	phase-averaged X -power, actual core
TPC-169	one actual core cell	Lebesgue L^2	all prefixes in a θ -shell	phase-averaged X -power, actual core
TPC-170	explicit packet corridor	Lebesgue-a.e. fixed phase	all prefixes, prescribed schedule	phase-metric X -power, actual core

Table 1: The arithmetic quantifiers do not reach a named physical atom or actual active support.

4 Architecture routes and arithmetic methods

The manifest has two disjoint route families. Complete architecture routes are

$$\mathcal{R}_{\text{arch}} = \{\text{archive}, \text{map}+, \text{scalar} + \text{ETO}\}. \quad (15)$$

The archive-only cell retains its scoped stop. The selected occurrence-augmented map route and the scalar-plus-ETO route remain NOT-TESTABLE. No theorem says that $\mathcal{R}_{\text{arch}}$ is the complete universe of possible architectures.

Arithmetic method subroutes include periodic major arcs, a direct fixed-atom twist, a source-backed metric-to-atom bridge, the uncontrolled atomic promotion and a pointwise bad-endpoint theorem. They may prove children of a complete architecture. They cannot be selected as replacement architectures and cannot justify an architecture-level reroute.

Proposition 4.1 (H9 arithmetic firewall). *The production occurrence, literal-weight, phase, endpoint and normalization registries have no direct or transitive arithmetic ancestor in the integration DAG.*

Proof. The H9 physical registry has as parents only the occurrence registry and the four independent physical registries. The occurrence registry descends only from the three structural roots in (5). Therefore its ancestor closure contains no node with arithmetic-core, arithmetic-target or arithmetic-negative role. Arithmetic nodes consume H9 records; they cannot manufacture them. $\square$

This direction matters for the phase-metric corridor. TPC-170 does not produce a physical phase registry. A future source-backed metric-to-fixed-atom theorem would consume both the TPC-170 theorem and a separately sourced phase record.

The four H9 leaves are data records, not decay theorems. Their decay coordinate is **NONE**. The literal-weight and normalization leaves carry their respective literal-data carrier axes; the phase leaf carries named-fixed-atom data; and the endpoint leaf carries deterministic-endpoint data. Only a later arithmetic estimate that consumes these records may carry a fixed- X decay coordinate.

5 Typed frontiers

Let $\mathcal{A}$ be the active ancestor closure of the selected map root. As before, minimal NOT-TESTABLE nodes suppress their unavailable descendants. The manifest recomputes the antichain from the DAG rather than trusting a submitted list.

Theorem 5.1 (Current selected-root antichain). *The selected map synthesis has minimal not-testable antichain*

$$\begin{aligned} \mathcal{B}_{\text{NT}} = \{ & \text{H1.source_backed_local_occurrence_edge_family}, \\ & \text{H1.actual_active_support_certificate}, \\ & \text{H1.canonical_minimal_representation_certificate}, \\ & \text{H9.literal_weight_registry}, \text{H9.phase_cell_registry}, \\ & \text{H9.endpoint_registry}, \text{H9.normalization_registry} \}. \end{aligned} \quad (16)$$

The canonical first reporting representative is the local occurrence-edge family. Separately, the parent-ready open frontier is

$$\mathcal{F}_{\text{open}} = \{\text{O161.direct_additive_twist_fixed_atom}, \text{O161.bad_endpoint_pointwise_fixed_atom}\}. \quad (17)$$

Proof. The three H1 roots are independent by [proposition 1.2](#). The four H9 registries are parent-free production targets and do not descend from the H1 roots. Every other unavailable node in the selected root descends from at least one of these seven. This proves (16). The two nodes in (17) are status OPEN with no unavailable active parent. Their status and type are therefore disjoint from the NOT-TESTABLE antichain. $\square$

6 Endpoint Ledger V4

Endpoint Ledger V4 preserves the six literal questions from V3: literal physical coefficients, fixed physical h_0 , physical atomic normalization, canonical/minimal representation, actual active support and strict 1/400 slack. Only fixed $h_0 = 2$ is currently proved in the relevant actual core.

V4 adds a registry gate and records the phase mode. The proved mode is

$$\text{LEBESGUE_AE_FIXED_PHASE};$$

the required physical mode is

$$\text{NAMED_FIXED_ATOM}. \quad (18)$$

The metric exponent in (11) is entered in a separate nonpromotable charge. Hence the named-fixed-atom exponent remains

$$\sigma_{\text{atom}} = 0, \quad \sigma_{\text{required}} = \frac{1}{400}. \quad (19)$$

Unknown registry costs are not zero, and no charge may be owned twice. The full synthesis has no defined net slack.

Theorem 6.1 (TPC-171 integration verdict). *For the source-locked TPC-163–170 state,*

$\begin{aligned} \text{Verdict}_{163:170} &= \text{NOT-TESTABLE}, \\ \text{FirstMissing} &= \text{H1.source_backed_local_} \\ &\quad \text{occurrence_edge_family}. \end{aligned}$

(20)

The full seven-node blocker antichain is retained. The arithmetic advance is L1 actual-core phase-metric all-prefix packet control; the program-positive L2 flag remains false.

Proof. The source hashes validate integrity but are not theorem evidence. The selected architecture has no exact stop and its route universe is not proved complete. Its active DAG has the nonempty antichain (16), so its predicates are not yet testable on the actual carrier. Independently, [theorem 2.1](#) has the phase quantifier on the left of the forbidden promotion (14), while the literal phase, support, normalization and endpoint registries remain unavailable. Thus neither a named-atom packet saving nor strict endpoint slack can be entered in the physical ledger. The declared selection order chooses the source-backed local edge family without collapsing the antichain. $\square$

7 Reproducible audit and next objects

The standard-library generator locks every TPC-163–170 main source, schema, script and machine-readable output under canonical UTF-8/LF hashing. It validates the census constants, the unique archive key, the three TPC-165 production gates, the TPC-166 root antichain and the claim boundaries of all four arithmetic papers. It then recomputes DAG ancestry, both frontiers, the H9 firewall and Endpoint Ledger V4.

Mutation regressions reject source-hash proof semantics; zero frozen edges promoted to actual nonexistence; archive addresses promoted to occurrence identities; formal support promoted to active

support; Lebesgue-a.e. phase promoted to a named atom; arithmetic ancestry in H9; an arithmetic method inserted into the architecture universe; unproved route-universe completeness; collapsed or merged frontiers; metric power charged as named-atom power; and a 1/400 pass with unknown literal fields.

The next structural object is one theorem-backed local edge family with exact source locators and compatible overlaps. The two parallel certificates in (5) remain independently necessary. The arithmetic choice is either a source-backed bridge placing the production phase outside the schedule null set, or a genuinely pointwise fixed-atom theorem. Neither outcome is assumed here.

8 Conclusion

This batch advances the arithmetic theorem from almost endpoints to all prefixes along a prescribed phase-metric packet corridor, and it resolves the structural monolith into three exact roots. It also locates the next hard quantifier change: phase-metric fixed powers do not pay a named physical atom. The path remains open, but the current state is still NOT-TESTABLE, with no positive program-L2, 1/400, prime-pair or twin-prime conclusion.

References

- [1] Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, fifth edition, 2019.
- [2] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory*. Cambridge University Press, 2007.
- [3] Liang Wang. A Source-Locator Census for the Occurrence Crosswalk: Native-Key Collisions and the Zero-Edge Boundary, 2026. TPC-163 manuscript.
- [4] Liang Wang. The Minimal Archived Separation Key: Exact Row Addressing without Occurrence Semantics, 2026. TPC-164 manuscript.
- [5] Liang Wang. Source-Backed Local-to-Global Crosswalk Gluing: A Formal Descent Theorem and Three Production Gates, 2026. TPC-165 manuscript.
- [6] Liang Wang. A Refined H1 Crosswalk Frontier: Three Independent Production Roots, 2026. TPC-166 manuscript.
- [7] Liang Wang. A Parseval Corridor for Direct Additive Twists on the Determinant-Two Möbius Core, 2026. TPC-167 manuscript.
- [8] Liang Wang. Separated Phase Registries for Direct Core Twists: A Finite Large-Sieve Gate and a Selector Firewall, 2026. TPC-168 manuscript.
- [9] Liang Wang. One Phase-Exceptional Set for Every Atomic Prefix: A Dyadic Maximal Parseval Theorem on Determinant-Two Fibers, 2026. TPC-169 manuscript.
- [10] Liang Wang. Metric Packet-Corridor Return: Borel–Cantelli, Abel Summation, and the Fixed-Atom Stop, 2026. TPC-170 manuscript.
- [11] Antoni Zygmund. *Trigonometric Series*. Cambridge University Press, third edition, 2002.